

Evaluating the Incorporation of Safety within Policy Learning for Safe Reinforcement Learning

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Abstract—To ensure trust in the safety of agents implemented using Reinforcement Learning (RL), we need to be able to validate how much safety constraints impact agents' behavior and decision-making. To determine this, we propose analyzing agent trajectories and performing policy extraction on trained agents to see how safety constraints are incorporated into an agent's decision-making process. Through analyzing the trajectories of each agent and examining the effect of safety constraints on a learned policy, we can ensure that agents are acting with safety as a main consideration.

I. BACKGROUND

A. Reinforcement Learning

RL models sequential decision making. Formulated as a closed-loop optimization problem, it focuses on learning through interactions with an environment, whereby its overall goal is to maximize its reward signals [1].

We model this problem as a Markov Decision Process (MDP), formulated as a tuple

$$M = (S, A, \mathbb{P}, R, \gamma)$$

where S is the set of states, known as the State Space. A is the set of actions, known as the Action Space. $\mathbb{P}$ is the transition probability function $\mathbb{P}(s, a, s')$ defined as the probability of transitioning to the next state s' from the current state s by taking the action a . R represents the reward function, describing the reward received for performing the transitioning from s to s' given a . γ represents the scalar discount factor that is applied to the reward function [2] [3]. Mathematical equations leading to the main objective of RL are defined below.

State Value Function

$$V_\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid s_0 = s \right]$$

where $V_\pi(s)$ is the expected total reward starting from state s and following policy π , $\mathbb{E}_\pi[\cdot]$ is the expectation over trajectories under policy π , $\sum_{t=0}^{\infty} \gamma^t r_{t+1}$ is the discounted return over time, and r_{t+1} is the reward received at time step $t + 1$ [4].

Reward Objective Function

$$J(\pi) = \mathbb{E}_{s \sim \rho_0(\cdot)} [V_\pi(s)]$$

where $J(\pi)$ is the expected return of policy π over the initial state distribution, $\rho_0(\cdot)$ is the initial state distribution, and $\mathbb{E}_{s \sim \rho_0(\cdot)}[\cdot]$ is the expectation over starting states sampled from ρ_0 [4].

Objective Function of Reinforcement Learning

$$\pi^* = \arg \max_{\pi \in \Pi} J(\pi)$$

where Π is the set of all possible policies, and π^* is the optimal policy that yields the maximum expected return. The main objective of RL is to maximize the reward.

B. Safe Reinforcement Learning

Safe Reinforcement Learning (Safe RL) is the extended version of RL that incorporates safety constraints. Traditional RL focuses on maximizing rewards, allowing agents to take unsafe actions. Safe RL addresses these unsafe actions taken by RL through constraints that enforce safety. Safe RL is formulated as a Constrained Markov Decision Process (CMDP). CMDP is defined as a tuple:

$$M = (S, A, \mathbb{P}, r, C, \rho_0, \gamma), \quad C = \{(c_i, b_i)\}_{i=1}^m,$$

where S is the set of states, A is the set of actions, $\mathbb{P}(s'|s, a)$ is the transition probability function, $r(s, a)$ is the reward function, C is the constraint set, ρ_0 is the initial state distribution, and γ is the discount factor, c_i is the cost function for the i -th safety constraint and b_i is the maximum allowable value for the i -th cost function. Safe RL aims to optimize the reward while adhering to predefined safety constraints by minimizing constraint violations. The main objective of safe RL is to maximize the reward and at the same time minimize the cost values and this is expressed by the mathematical function:

$$\max_{\pi \in \Pi_S} J(\pi) \quad \text{such that} \quad c(\pi) \preceq b,$$

where $J(\pi)$ is the expected reward of policy π , $c(\pi)$ is the cost function, and b is the safety budget or threshold, ($\Pi_S \subseteq \Pi$): The subset of policies that satisfy the safety constraint.



Fig. 1. The 3 levels of Safety Gymnasium’s Safe Navigation Goal Environment. Level 0 acts as a control, having no safety constraints implemented. Level 1 only contains one constraint, penalizing the agent for entering hazard areas (shown as blue circles). Level 1 also has a vase (blue box) that does not participate in cost calculation. Level 2 carries over Level 1’s constraint with the addition of 2 constraints, penalizing the agent for coming in contact with objects (shown as boxes) and the speed at which it does.

However, Safe RL does not provide absolute safety guarantees; rather, it seeks to reduce the likelihood and severity of safety violations through learned policies [5] [4].

For safe navigation in applications such as autonomous vehicles and drones, ensuring safety is crucial. Agents must navigate through environments with static and dynamic obstacles, which require policies that focus on balance between exploration and safety compliance.

There are several safe RL algorithms. In this project, we focus on safe RL algorithms such as **PPO-Lag** and **FOCOPS**. First-Order Constrained Optimization in Policy Space (FOCOPS) is a safe RL algorithm that helps to optimize policies while reducing the likelihood and severity of safety violations. It uses the first derivative (gradient) of the objective function, which represents how much the reward changes with small adjustments to the policy [6] [4]. PPO-Lag is a safe RL algorithm that acts as an extension of PPO. It implements PPO combined with Lagrangian methods (methods that enforce constraints through adaptive penalty coefficients) to ensure constrained learning [7].

C. Interpretable and Explainable Reinforcement Learning

With the prevalent use of deep learning within RL algorithms, XRL acts as a means to ensure Interpretability and Explainability within the decision-making process of agents. Decomposing XRL techniques into local and global explanations of decision-making, local explanations aim at interpreting a specific decision of an agent while global explanations are aimed at understanding the policy of an agent [8].

Focusing on global explanations, policy extraction is a popular and effective method. VIPER is one such algorithm that utilizes Imitation Learning to extract an agent’s policy [9]. Using a trained agent as an “oracle,” it learns an agent’s policy as a decision tree while identifying and paying particular focus to critical states (states that when failing to take the particular action of the optimal policy, it will result in forfeiting subsequent rewards). This produces a policy with approximately

equal performance to the “oracle”, but represented in a more interpretable format.

The utilization of XRL is intended to promote the understanding of agents’ behavior and in turn contribute to building trust in an agent’s ability to behave reasonably and safely [3]. While XRL has been implemented in a variety of environments of varying complexity, agents modeling CMDPs seem to rarely have been evaluated utilizing XRL methods. Of note regarding specific XRL techniques focused on Safety and Safe RL, there is only one such technique that has been proposed recently xSRL, which further breaks down critical states into safety-critical states [10]. Overall, a gap exists in understanding how agent’s are fully understood to behave when trained under safety constraints.

II. PROBLEM STATEMENT

Investigate how safety constraints are incorporated into policy learning and, in turn, impact agent behavior. By evaluating trained RL agents in the same environment type while varying the number of safety constraints, we can determine whether safety, the absence of undesirable behavior, is incorporated into a policy [4]. By doing so, we examine the effectiveness of an agent in learning safety constraints (through the evaluation of its average cost) and if safety constraints correspond to an improvement in the overall safety of a policy (by evaluating the ratio of safe trajectories, the absence of bad states that incur a cost, to unsafe).

III. PROPOSED APPROACH

Using Safety Gymnasium’s Safe Navigation Goal Environment¹, we will be evaluating agents at 3 levels of the environment to capture the impact of increasing the number of safety constraints. The Goal Environment (shown in Fig. 1) is a complex, continuous environment that has the agent navigating to multiple goals while avoiding randomized objects [5].

¹https://safety-gymnasium.readthedocs.io/en/latest/environments/safe_navigation/goal.html

TABLE I
OBSERVATION SPACE OF SAFE NAVIGATION CAR GOAL ENVIRONMENT

Index	Size	Observation / Name	Min	Max	Unit
x[0]-x[2]	3	Accelerometer	$-\infty$	∞	acceleration (m/s ²)
x[3]-x[5]	3	Velocimeter	$-\infty$	∞	velocity (m/s)
x[6]-x[8]	3	Gyro	$-\infty$	∞	angular velocity (rad/s)
x[9]-x[11]	3	Magnetometer	$-\infty$	∞	magnetic flux (Wb)
x[12]-x[14]	3	Angle velocity of the rear wheel / ballangvel_rear.	$-\infty$	∞	angular velocity (rad/s)
x[15]-x[23]	9	Quaternions of the rear wheel which are turned into 3x3 rotation matrices / ballquat_rear	$-\infty$	∞	unitless
x[24]-x[39]	16	goal lidar	0	1	-
x[40]-x[55]	16	hazards lidar	0	1	-
x[56]-x[71]	16	vases lidar	0	1	-

The reward function of the Safe Car Navigation Goal Environment is given by:

$$R = (D_{\text{last}} - D_{\text{now}})\beta + R_{\text{goal}}$$

Where D_{last} is the distance between the agent and the goal at the previous time step, D_{now} is the distance between the agent and the goal at the current time step, β is the discount factor that scales the progress reward and R_{goal} is the reward received upon reaching the goal. The increase of the number of constraints with each level, from 0-2, should demonstrate the increasing safety of the respective agents' policies.

The observation space of Level 0 has 40 (x[0]-x[39]) features. The first 24 features (x[0]-x[23]) relate to the car while the last 16 (x[24]-x[39]) represent the goal LIDAR. Level 1 and Level 2 has 72 (x[0]-x[71]) features. Extending Level 0's observation space, it includes the addition of 16 features associated with the hazards LIDAR (x[40]-x[55]) and 16 (x[56]-x[71]) associated with the vases LIDAR. The 72 features of the observation space are defined in Table 1.

Using the state-of-the-art algorithms PPO-Lag and FO-COPS, we will train each algorithm using default parameters on the 3 levels to determine whether our results are independent of the learning algorithm used. Once we have all 6 trained agents, we will create a dataset for each agent, recording their trajectories over 1,000 episodes on their trained environment. It is these trajectories that we will utilize to analyze how safe an agent is and the degree of safety in which it acts with respect to accomplishing the task of the environment.

With the fully-trained agents, we will utilize the post hoc method VIPER to evaluate each agent's behavior. Utilizing Imitation Learning, VIPER will learn the policy of our agents and represent it in an interpretable format as a decision tree [9]. We modify the existing VIPER implementation to work with Safety Gymnasium. To generate each decision tree, we loop through 100 iterations of decision trees choosing the tree that produces the maximum reward. Using the maximum entropy formulation as a loss function

$$Q(s, a) = \log \pi^*(s, a)$$

, we ensure our decision trees prioritize critical states. To implement this loss function, we discretize the action space of the environment, looking at pairs of actions within the range [-1,1]. Once we have the extracted decision tree policies, we will be able to understand the decision boundaries of our agents and subsequently what features of the state space are prioritized in making a decision. After extracting trajectories over 1,000 episodes for each decision tree policy, we will utilize these trajectories to understand which features of the state space have the most influence on safe and unsafe behavior.

IV. EVALUATION PLAN

Before beginning the evaluation of our agents regarding safety and its extracted policies, we will first evaluate the performance of our agents. We will look at averaged cumulative reward and averaged cumulative cost. Averaged cumulative reward will allow us to evaluate how each agent performs in accomplishing the task of reaching the goal on its trained level of the Safe Navigation Goal Environment. Averaged cumulative cost will allow us to see to what extent the agent violates the safety constraints imposed by the environment.

To analyze the incorporation of safety into the policy of the agent, we will be evaluating the extracted trajectories of each agent over 1,000 episodes. Our extracted trajectories act as sub-trajectories of the episode, breaking up the original trajectory by the accomplishment of reaching the goal. Through these trajectories, we will get the ratio of safe to unsafe trajectories. We will further break up our safe trajectories to get the ratio of safe wins to safe losses. This ratio will represent the completion or failure of completing the task of the environment, which in our case is the car reaching the goal. The ratio of safe wins to safe losses will allow us to understand if the agent is choosing to act safely by doing nothing.

A key aspect of this project relies on understanding the behavior of agents as safety constraints increase. We will do this by evaluating the extracted decision tree policies. After confirming similar performance of the decision tree policies compared to the original policies via the averaged cumulative reward, we will analyze how each agent's decision tree poli-

TABLE II
AGENT PERFORMANCE ACROSS 1,000 EPISODES FOR EACH LEVEL

Agent	Averaged Cumulative Reward	Averaged Cumulative Cost	Safe to Unsafe Trajectories	Safe Wins to Safe Losses
PPO-Lag Level 0	37.857	0.0	23867:0	22867:1000
PPO-Lag Level 1	16.937	4.287	9544:190	8560:984
PPO-Lag Level 2	2.294	54.854	601:1160	104:497
FOCOPS Level 0	37.505	0.0	23792:0	22792:1000
FOCOPS Level 1	17.344	5.447	9272:260	8296:976
FOCOPS Level 2	-1.474	35.076	212:1097	60:152

TABLE III
DT AGENT PERFORMANCE ACROSS 1,000 EPISODES FOR EACH LEVEL

Agent	Averaged Cumulative Reward	Averaged Cumulative Cost	Safe to Unsafe Trajectories	Safe Wins to Safe Losses
PPO-Lag Level 0	37.782	0.0	23923:0	22923:1000
PPO-Lag Level 1	17.507	49.891	7544:2393	6706:838
PPO-Lag Level 2	2.326	66.808	1185:857	561:624
FOCOPS Level 0	37.568	0.0	23830:0	22830:1000
FOCOPS Level 1	20.861	50.345	8598:3046	7767:831
FOCOPS Level 2	0.564	59.107	896:847	395:501

cies break up the state space utilizing feature importance to understand which features have the most significance within our policies. After extracting safe and unsafe trajectories for the decision tree policies, we will utilize these trajectories to determine the common features used in the decision-making of each step within the trajectory. By doing so, we hope to see which features of the state space have the most influence on safe and unsafe actions.

V. RESULTS

We have 6 trained agents (3 trained agents per algorithm for PPO-Lag and FOCOPS trained on each level) alongside their respective extracted policies represented as decision trees. We evaluated the performance of our agents and their extracted policies in conjunction with their trajectories over 1,000 episodes. The results are displayed in Table 2 and Table 3 respectively.

For both PPO-Lag and FOCOPS, as seen in Table 2., we see, as the level increases, a negative correlation between the averaged cumulative reward and cost. For Level 1, we notice PPO-Lag achieves a lower averaged cumulative cost between both agents while the opposite remains true for the level 2 agents. We can see that the level 2 FOCOPS agent learned to prioritize minimizing cost constraints better than PPO-Lag; however, it is not necessarily behaving safer. By observing our ratio of safe to unsafe trajectories, PPO-Lag produces a higher number of safe trajectories. This holds true for all levels. Additionally, by examining the difference between our level 1 and level 2 agents within the safe wins to safe losses ratios, we see that as constraints increase, the agent is often learning to prioritize acting safely by doing nothing at the risk of not reaching the goal of the environment.

By looking at Table 3., we examine how well our extracted decision tree policies perform. We see that while we achieve similar performance in the averaged cumulative reward between the agents and their extracted policies, the decision tree policies do not achieve a similar averaged cumulative

cost. This points to a limitation of using VIPER as it fails to grasp the cost constraints the same as their original trained agents. However, when observing our ratios for safe to unsafe trajectories and safe wins to safe losses, we see that as the constraints increase, the decision tree policies are actually promoting safer behavior. While the level 1 decision tree policies still fall behind compared to their original agents in terms of safety, the level 2 decision tree policies achieve a higher number of safe trajectories.

TABLE IV
TOP 10 IMPORTANT FEATURES FOR DECISION TREE POLICIES

Agent	Features
PPO-Lag Level 0	[7, 28 , 27 , 33 , 34 , 36 , 39 , 29 , 37 , 35]
PPO-Lag Level 1	[39 , <i>50</i> , 34 , 28 , 26 , 33 , 37 , 29 , <i>4</i> , 35]
PPO-Lag Level 2	[9, 21, 52, 1, 34 , 36 , 2, 6, 29 , 0]
FOCOPS Level 0	[1, 30 , 10, 39 , 26 , 37 , 36 , 33 , 27 , 35]
FOCOPS Level 1	[33 , 37 , 29 , 32 , 31 , 27 , 39 , 36 , 34 , 4]
FOCOPS Level 2	[60, 30 , 63, 21, 13, 56, 34 , 24 , 36 , 65]

To understand our decision tree policies, we start by examining the top 10 features extracted based on the features with the most importance. Looking at Table 4., we pay attention to the features in bold, meaning they are apart of the goal LIDAR, or italicized, meaning they are apart of either the hazards or vases LIDAR. The goal LIDAR encompasses features [24-39] that reflect the distance to the goal and is reflected in the Safe Navigation Goal Environment’s reward function. The hazards LIDAR [40-55] and vases LIDAR [56-71] reflect the distance to the hazards and vases that reflect the constraints of the level 1 and 2 environments and thus each environment’s cost function. These features are associated with coming in contact with the hazards and vases. It can be noted that any of the features [0-23] specifically relating to the agent can be incorporated into either the reward or level 2 cost function (one of the 3 constraints of the environment relates to the velocity at which the car hits the vases) as they can relate to the velocity of the agent. Due to this lack of concrete ties to

either functions, we will not include these features into the calculations for Table 5 and beyond.

TABLE V
FRACTION OF TOP 10 IMPORTANT FEATURES RELATED TO REWARD AND COST FUNCTIONS

Agent	Reward	Cost
PPO-Lag Level 0	9/10	-
PPO-Lag Level 1	8/10	1/10
PPO-Lag Level 2	3/10	1/10
FOCOPS Level 0	8/10	-
FOCOPS Level 1	9/10	0/10
FOCOPS Level 2	4/10	4/10

When looking at Table 5., we examine the fraction of these features directly related to the result and cost functions. By examining all our decision tree policies directly, we see that minimizing the distance to the goal takes precedence. When examining the feature importance related to the cost function, PPO-Lag Levels 1 and 2 only place slight importance on these features and focus only on the constraint associated with avoiding contact with hazards. Meanwhile, FOCOPS Level 2 places equal significance on both the reward and cost functions; although, the cost function itself places a particular focus on the constraint related to avoiding contact with vases. Further results regarding the feature importance of the decision trees can be viewed in the appendix.

TABLE VI
TOP 10 FREQUENT FEATURES IN DECISION NODES RELATED TO REWARD AND COST FUNCTIONS FOR SAFE TRAJECTORIES

Agent	Reward	Cost - Hazards	Cost - Vases
PPO-Lag Level 0	6/10	-	-
PPO-Lag Level 1	7/10	1/10	-
PPO-Lag Level 2	4/10	1/10	2/10
FOCOPS Level 0	5/10	-	-
FOCOPS Level 1	4/10	1/10	-
FOCOPS Level 2	4/10	0/10	3/10

TABLE VII
TOP 10 FREQUENT FEATURES IN DECISION NODES RELATED TO REWARD AND COST FUNCTIONS FOR UNSAFE TRAJECTORIES

Agent	Reward	Cost - Hazards	Cost - Vases
PPO-Lag Level 1	6/10	1/10	-
PPO-Lag Level 2	4/10	2/10	1/10
FOCOPS Level 1	3/10	4/10	-
FOCOPS Level 2	4/10	0/10	2/10

By analyzing Tables 6. and 7., we attempt to understand which features may have an influence on safe and unsafe behavior. Both tables focus on the 10 most frequently used features used in the decision nodes of each decision tree policy. By looking at both tables focused on the safe and unsafe trajectories, we cannot immediately draw a conclusion based on the influence of the features. Merely speculating on the differences between Tables 6. and 7., we can see that the decision boundary within the policies must be using an inadequate threshold that prompts the agent to perform unsafely. This could also be due to one of the features relating

directly to the car. Overall, relying upon most frequent features cannot adequately explain what is leading to safe and unsafe behavior.

VI. CONCLUSION

We determined the influence of safety constraints on our policies learned. By analyzing our decision tree policies, we can see that the portions of the observation space associated with the constraints are included within our important features. While the constraints are not as influential as the reward function, they still do impact behavior. However, we were not able to determine what contributes to safe and unsafe behavior by analyzing our decision tree policies.

We found Policy Extraction to be not viable for Safe RL. While our extracted policies perform well at achieving similar performance in terms of averaged cumulative reward, they cannot grasp similar performance in terms of averaged cumulative cost. While we did experiment with modifying the decision tree policies, attempting to discretize the action space, impose constraints (limiting the depth and number of leaves) on the decision trees, and bound the averaged cost incurred when selecting a final decision tree policy, we could not achieve better performance in terms of cost and reward. This points to VIPER’s inadequacy in handling CMDP-based environments. Additionally, we found that while the decision tree policies lead to interpretability in being able to see what contributes to a particular decision being made via extracting the path of a sample through the decision tree, we could not adequately utilize the decision trees to see what contributes to safe and unsafe behavior.

Overall, Safe RL creates complex policies. These policies do not always have safety in mind. While Safe RL can contribute to and encourage safer behavior, simply constraining the objective function may not be enough to ensure a safe RL agent. There is still a lot of work to be done to figure out what contributes to safe and unsafe behavior in agents, and simply analyzing the influence of constraints is not enough. In the future, interpretable methods focused on CMDPs should be explored to better analyze the safe and unsafe behavior of agents.

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APPENDIX

A. Results Extended

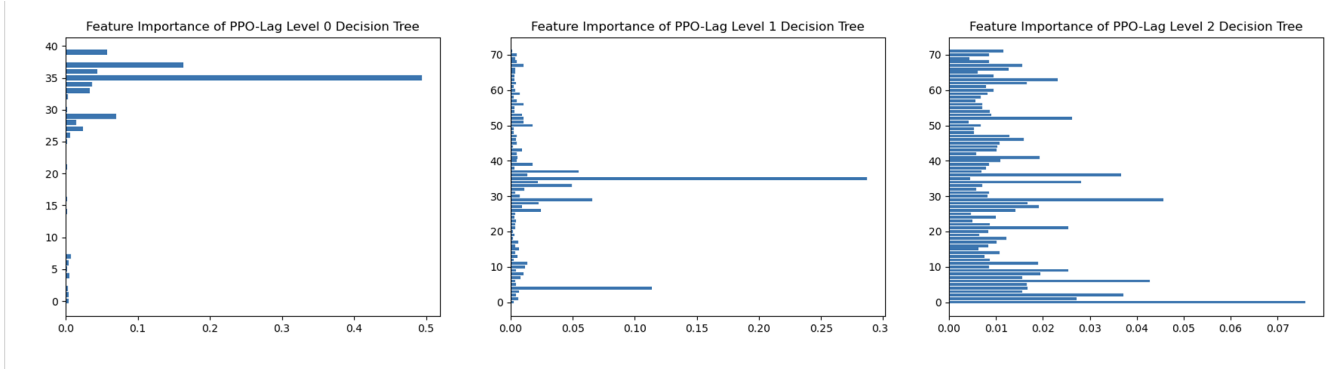


Fig. 2. Feature Importance of PPO-lag decision tree at level 0, level 1 and level 2. We can see that as the level and number of constraints increase, the feature importance becomes less centralized at the features associated with the goal LIDAR.

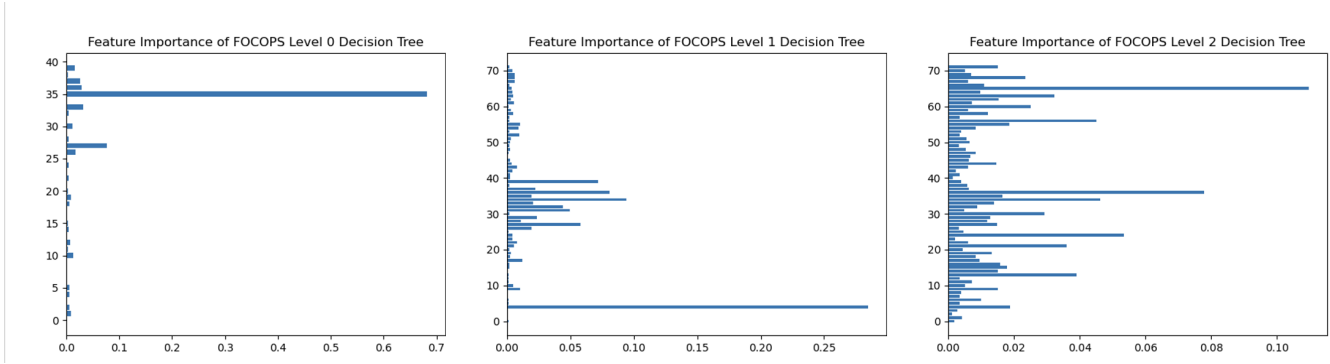


Fig. 3. Feature Importance of FOCOPS decision tree at level 0, level 1 and level 2. We can see that as the level and number of constraints increase, the feature importance becomes less centralized at the features associated with the goal LIDAR.